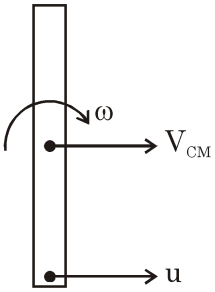
SOLUTION
SECTION-I

Single Correct Answer Type **5 Q. [3 M (–1)]**

1. Ans.(C)

Sol.



$$V_{CM} - \frac{\omega L}{2} = u$$

$$V_{CM} = u + \frac{\omega L}{2}$$

$$KE = \frac{1}{2}MV_{CM}^2 + \frac{1}{2}I_{CM}\omega^2$$

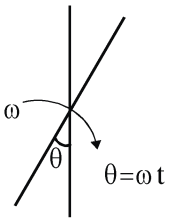
2. Ans.(C)

Sol. $(dS)^2 = (dx)^2 + (dy)^2 \Rightarrow dS = \sqrt{\left(\frac{dy}{dx}\right)^2 + 1}\, dx$ ...(i)

Velocity of B after time t.

$$\frac{dx}{dt} = v_x = \left(\frac{\omega\ell}{2} - \frac{\omega\ell}{2}\cos\omega t\right) \text{...(ii)}$$

$$\frac{dy}{dt} = v_y = \left(\frac{\omega\ell}{2}\sin\omega t\right) \text{...(iii)}$$



From (ii) and (iii) $\Rightarrow \frac{dy}{dx} = \frac{\sin\omega t}{1-\cos\omega t}$

Now $dS = \sqrt{\left(\frac{\sin\omega t}{1-\cos\omega t}\right)^2 + 1}\ \text{dx}$; $dx = v_x dt$

$$dS = \sqrt{\left(\frac{\sin\omega t}{1-\cos\omega t}\right)^2 + 1}\ \frac{\omega\ell}{2}(1-\cos\omega t)\, dt$$

$$dS = \frac{\omega\ell}{2}\sqrt{(\sin\omega t)^2 + (1-\cos\omega t)^2}\ \text{dt}$$

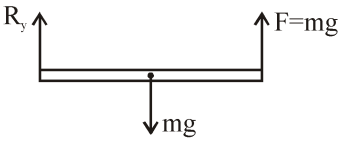
$$dS = \frac{\omega\ell}{2}\sqrt{\sin^2\omega t + \cos^2\omega t + 1 - 2\cos\omega t}$$

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$$\Rightarrow dS = \frac{\omega\ell}{2}\left(2\sin\frac{\omega t}{2}\right) \Rightarrow \int_0^S dS = \omega\ell \int_0^{T=\frac{2\pi}{\omega}} \sin\frac{\omega t}{2}\ dt = \frac{\ell\omega}{\left(\frac{\omega}{2}\right)}\left[-\cos\frac{\omega t}{2}\right]_0^{\frac{2\pi}{\omega}} = 4\ell$$

3. Ans.(C)

Sol.



$$mg\ell - mg\frac{\ell}{2} = \frac{m\ell^2}{3}\alpha \qquad \text{... (i)}$$

$$\alpha = \frac{3g}{2\ell}$$

$$R_y + F - mg = m\frac{\ell}{2}\alpha \qquad \text{... (ii)}$$

$$R_y = m\frac{\ell}{2}\times\frac{3g}{2\ell} = 15\ \text{N}$$

4. Ans.(D)

Sol. Torque is non-zero but work done by torque is zero.
Here torque is opposite to angular momentum due to which angular momentum of ball decreases.

5. Ans. (C)

Sol. $\vec{L} = \vec{r} \times \vec{p}$

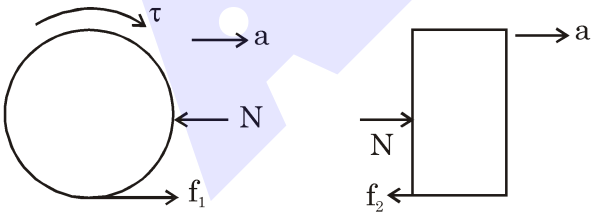
$$= \left[\sqrt{\ell - R^2}\ (-\hat{k}) + R\hat{i}\right] \times \left[Mv\,\hat{j}\right]$$

$$= Mvl\left[\frac{\sqrt{l^2 - R^2}}{l}\,\hat{i} + \frac{R}{l}\,\hat{k}\right]$$

Multiple Correct Answer Type — 8 Q. [4 M (–1)]

6. Ans. (C, D)

Sol.



$f_1 - N = 6a$
$N - f_2 = 3a$
$f_1 - f_2 = 9a$

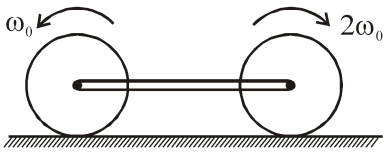
$$a = \frac{f_1 - f_2}{9} = 1\ \text{m/s}^2$$

$I = 0.12\ \text{kgm}^2$
$\tau - f_1 R = I\alpha$
$\alpha = 10\ \text{rad/s}^2$

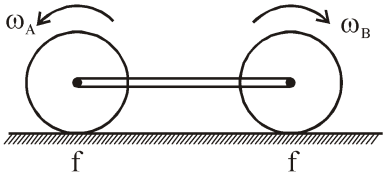
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7. Ans.(A, B, D)

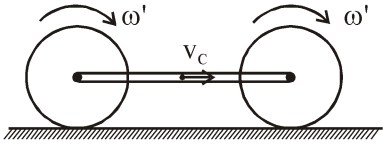
Sol.



Initially



at any instant 't'



Finally pure rolling takes place; $v_C = \omega' R$

Angular momentum conservation gives

$$(2\omega_0)\frac{mR^2}{2} - \frac{mR^2}{2}(\omega_0) = 2mv_C R + \frac{2mR^2}{2}\omega'$$

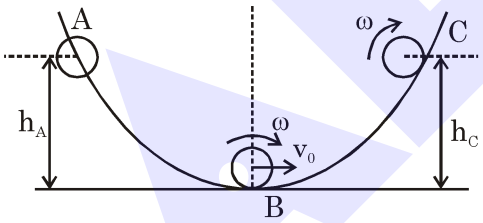
$$\frac{mR^2}{2}\omega_0 = 2mv_C R + mRv_C \quad (\text{Since } v_C = \omega' R)$$

$$v_C = \frac{R\omega_0}{6}$$

* Second figure shows that rod is under stress & hence its length will increase.
* Finally when the discs roll purely, there is no friction & hence no tensile stress
* Since friction is kinetic therefore mechanical energy will decrease.

8. Ans.(A,B)

Sol.



$$mgh_A = K_B = (K_{trans})_B + (K_{rot})_B$$

$$mgh_A = \frac{1}{2}mv_0^2 + \frac{1}{2}I_0\omega^2 = K_B \qquad \ldots (i)$$

$$\frac{1}{2}mv_0^2 + \frac{1}{2}I_0\omega^2 = \frac{1}{2}I_0\omega^2 + mgh_C, \text{ where } \frac{1}{2}I_0\omega^2 = K_C$$

$$mgh_C = \frac{1}{2}mv_0^2 \qquad \ldots (ii)$$

From (i) & (ii)

$h_A > h_C$; $K_B > K_C$

and $h_A > h_C$; $K_C > K_A$

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9. **Ans. (C,D)**

10. **Ans.(B, D)**

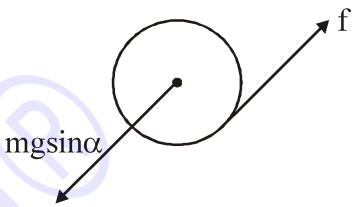
11. **Ans.(A, B, C)**

Sol. For (A) $a = \frac{1}{M}\left(Mg\sin\alpha - \frac{Mg\sin\alpha}{3}\right) = \frac{2}{3}g\sin\alpha$

For (B) $V^2 = 0 + 2 \times \frac{2}{3}g\sin\alpha\, l \Rightarrow V = \sqrt{\frac{4}{3}g l\sin\alpha} \Rightarrow \omega = \frac{v}{R} = \sqrt{\frac{4gl\sin\alpha}{3R^2}}$

For (C & D)

For pure roling $a = \alpha R$

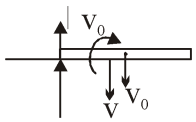


$$\frac{Mg\sin\alpha - f}{M} = \frac{f \times R^2}{\frac{MR^2}{2}} \Rightarrow Mg\sin\alpha - f = 2f \Rightarrow f = \frac{Mg\sin\alpha}{3}$$

$$\mu mg\cos\alpha = \frac{mg\sin\alpha}{3} \Rightarrow \mu = \frac{\tan\alpha}{3}$$

12. **Ans. (C,D)**

Sol. $mv - mv_0 = \int N dt$



$$\frac{m\ell^2}{12}\omega = \frac{\ell}{2}\int N dt$$

$$v - (-ev_0) = \frac{\omega\ell}{2}$$

$$v + ev_0 = \frac{\omega\ell}{2}$$

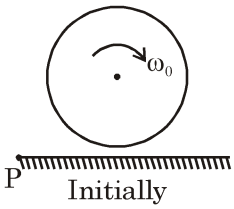
$$3 \times mv - mv_0 + \frac{m\omega\ell}{6} = 0$$

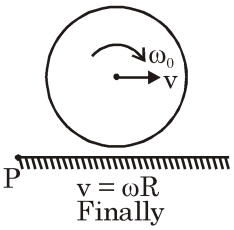
$$4v + (e-3)v_0 = 0$$

$$v = \frac{(3-e)v_0}{4}$$

13. **Ans.(A, B)**

Sol.





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$$\frac{mR^2}{2}\omega_0 = \frac{mR^2}{2}\left(\frac{v}{R}\right) + mvR$$

$$\frac{mR^2}{2}\omega_0 = \frac{3}{2}mvR$$

$$v = \frac{R\omega_0}{3}$$

$$\omega = \frac{v}{R} = \frac{\omega_0}{3}$$

$$\frac{R\omega_0}{3} = \mu g t \Rightarrow t = \frac{R\omega_0}{3\mu g}$$

$$s = ut + \frac{1}{2}at^2 = \frac{1}{2}\times\mu g\frac{R^2\omega_0^2}{9\mu^2 g^2} = \frac{R^2\omega_0^2}{18\mu g}$$

Linked Comprehension Type (4 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

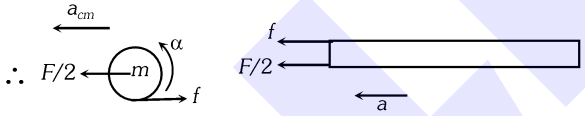
Paragraph for Question Nos. 14 to 16

14. Ans. (C)

Sol. $\frac{F}{2} - f = ma_{cm}$ ... (i) $fR = \frac{1}{2}mR^2\alpha$ ... (ii) $\frac{F}{2} + f = 2ma$ ... (iii)

For pure rolling $a_{cm} - R\alpha = a$ ...(iv)

On solving $f = \frac{F}{14}$, $a_{cm} = \frac{3F}{7m}$, $a = \frac{2F}{7m}$



15. Ans. (D)

16. Magnitude of friction force acting on the plank is
तख्ते पर कार्यरत् घर्षण बल का परिमाण होगा

(A) $\frac{F}{7}$ (B) $\frac{F}{14}$ (C) $\frac{F}{21}$ (D) $\frac{2F}{7}$

Ans. (B)

Paragraph for Question Nos. 17 to 19

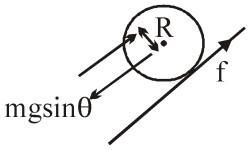
17. Ans.(D)

Sol. $F = mg\sin\theta - f$

$$\tau = 0 = Fh - fR \Rightarrow f = \frac{Fh}{R}$$

$$F = mg\sin\theta - \frac{Fh}{R}$$

$$F = \frac{mg\sin\theta}{1 + \frac{h}{R}}$$

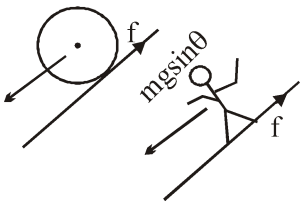


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F_{min} when 'h' is max = R

18. Ans. (A)

Sol. $a_s = \dfrac{g\sin\theta}{1+\dfrac{2}{5}\dfrac{mR^2}{mR^2}} = \dfrac{5}{7}g\sin\theta$



$ma_{sisyphus} = mg\sin\theta - f$

$f = \dfrac{2}{7}mg\sin\theta$ f is backwards

$mg\sin\theta - f_{stone} = ma_{stone}$

$f_{stone} = \dfrac{2}{7}mg\sin\theta$

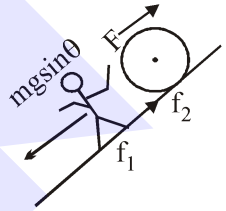
19. Ans. (A)

Sol. $F_{min} \Rightarrow F$ at R , $\alpha = 0$, $a = 0$

$mg\sin\theta + F = f_1$ $\rightarrow$ sisyphus

$F + f_2 = mg\sin\theta$ $\rightarrow$ stone

$FR = f_2R \rightarrow$ Stone



$\Rightarrow F = f_2$ $\Rightarrow F = \dfrac{mg\sin\theta}{2} = f_2 \le \mu mg\cos\theta$

$\mu \ge \dfrac{\tan\theta}{2}$

for sisyphus,

$mg\sin\theta + F = f_1 \le \mu mg\cos\theta$

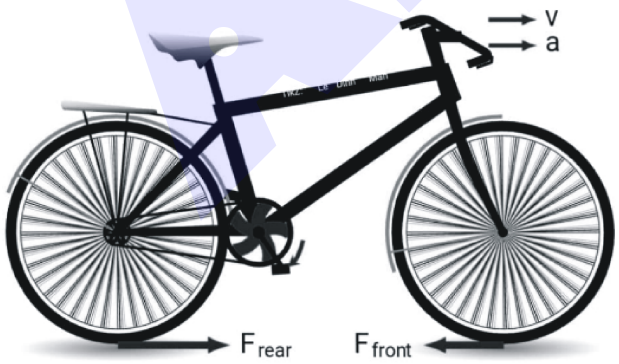
$\dfrac{3mg\sin\theta}{2} \le \mu mg\cos\theta$ $\Rightarrow \mu \ge 3/2\tan\theta$

For stone to roll as well as sisyphus to not slip $\mu \ge 3/2\tan\theta$.

Paragraph for Questions no. 20 to 22

20. Ans.(D)

Sol.



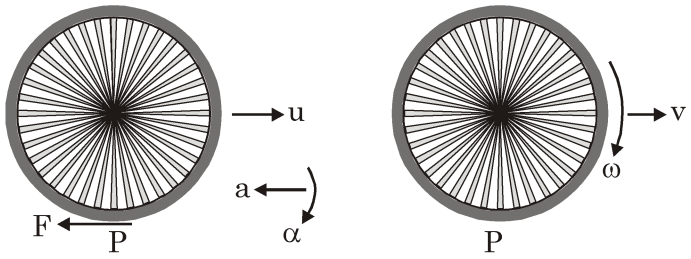
Friction Force on the Wheels during Pedaling

Friction Force on the Front Wheel

The front wheel is connected to the rest of the bicycle by a rod passing through its centre (axle). The

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torque on the wheel about its centre by the force coming from the rest of the bicycle is zero. Thus, pedaling can give linear velocity to the front wheel but cannot rotate it.



Motion of the Front Wheel on a Rough Horizontal Surface

Let us analyse a simple situation to get the direction of friction force on the front wheel. What will happen if a wheel translating with speed u is gently placed on a rough surface ?

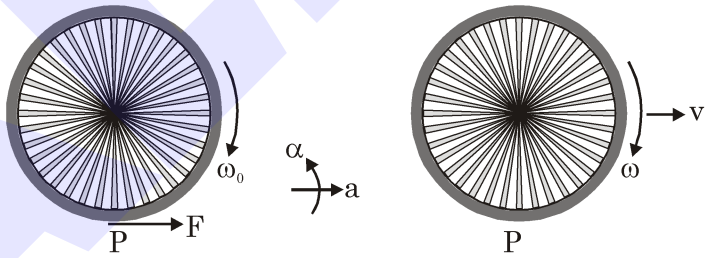
Every point on the wheel, including centre C and contact point P translates with the same speed u.

Thus, P moves towards right relative to the surface. To oppose this, frictional force at P acts towards the left. The frictional force retards the velocity of the centre. The torque about C due to frictional force gives clockwise angular acceleration and the wheel starts rotating in clockwise direction. If the coefficient of friction is sufficiently large then the wheel will start rolling without slipping after sometime. At this instant, velocity of P relative to the surface becomes zero. The frictional force will become zero and the wheel continues to roll without slipping, $\upsilon = \omega r$.

Friction Force on the Rear Wheel

The rear wheel is connected to rest of the bicycle by a rod passing through its centre and a chain connected to the pedals (see figure). Pressing the pedal increases tension in the upper portion of the chain.

This tension gives rise to clockwise torque and wheel starts rotating in clockwise direction.



Motion of Rear Wheel on a Rough horizontal surface during pedaling

Let us analyse a simple situation.

What will happen if a wheel rotating with angular velocity ω_0 is gently placed on a rough surface ?

Initial velocity of the contact point P is leftward relative to the surface. To oppose this, frictional force at P acts towards the right. The frictional force increases the velocity of C.

The torque about C due to frictional force is anti-clockwise. This torque gives anti-clockwise angular acceleration.

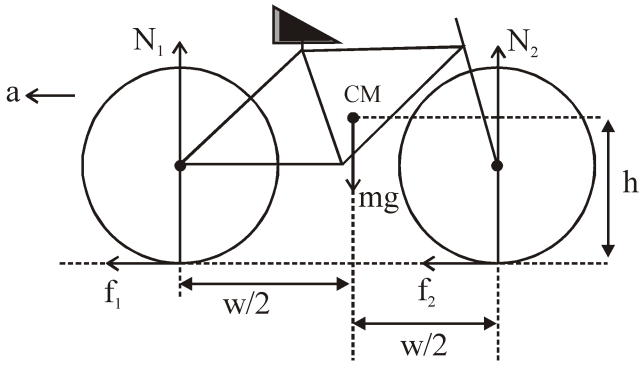
If the coefficient of friction is sufficiently large then the wheel will start rolling without slipping after some time. At this instant, velocity of P relative to the surface becomes zero. After this, the wheel continues to roll without slipping, $\upsilon = \omega r$.

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Case-2 (Q. 21 and 22) : The bicycle is moving to the right, but slowing down at a constant rate. The acceleration has a magnitude a. Assume that the coefficient of sliding friction between each tyre and the ground is μ, and that both tyres are skidding (sliding without rotating). Express your answers in terms of w, h, M and g.

21. Ans. (A)

Sol.



$f_1 + f_2 = f = ma$ ... (i)

$N = N_1 + N_2 = mg$ ... (ii)

$(\vec{\tau}_{net})_{cm} = 0$

$(f_1 + f_2)h + N_1 w - N_2 w = 0$ ... (iii)

$f_1 + f_2 = f = \mu N_1 + \mu N_2$ ... (iv)

From (ii) & (iii)

$$N_2 = \frac{1}{2}\left[mg - \frac{2f\,h}{w}\right]$$

$$N_1 = \frac{1}{2}\left[mg + \frac{2f\,h}{w}\right]$$

For both the tyres to remain in contact with the ground,

$N_2 \geq 0$

$$mg \geq \frac{2h}{w}(mg)\mu \quad \text{[using (iv)]}$$

$$\mu \leq \frac{w}{2h}$$

22. Ans.(B)

Sol. From (i), $f = ma$

$\mu mg = ma$

$a = \mu g$

$$\mu = \frac{a}{g}$$

$$\mu \leq \frac{w}{2h}$$

$$a \leq \frac{wg}{2h}$$

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Paragraph for Question Nos. 23 to 25

23. Ans.(C)

24. Ans.(A)

25. Ans.(A)

Sol. (23 to 25)

Let ball leaves with velocity u & disc rotates with ω clockwise direction

Angular moment conservation about centre of disc

$$\vec{L}_i = \vec{L}_f$$

$$O = muR - \left(\frac{MR^2}{2}\right)\omega$$

$$\omega = \frac{2mu}{MR} \qquad \text{... (i)}$$

Energy conservation

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 \qquad \text{... (ii)}$$

$$u = v\sqrt{\frac{M}{M+2m}}, \omega = 2\frac{mv}{MR}\sqrt{\frac{M}{M+2m}}$$

SECTION-II

Numerical Answer Type Question **1Q.[3(0)]**

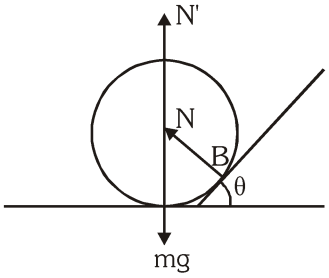
(upto second decimal place)

26. Ans. 44

Sol. Initial angular momentum of the solid sphere about B

$$\Rightarrow I\omega + mv_{cm}r = \frac{mR^2}{2}\frac{v_C}{R} + mv_C R\cos\theta$$

Final angular momentum of the solid sphere at B = $mvR + \frac{mR^2}{2}\frac{v}{R}$



Angular momentum about the B will remain conservation $L_i = L_F$

$$\frac{mR^2}{2}\cdot\frac{v_C}{R} + mv_C R\cos 60° = mvR + \frac{mR^2}{2}\frac{v}{R}$$

$$66\times\frac{R}{2} + 66\times\frac{R}{2} = v\times R + \frac{vR}{2}$$

$$66R = \frac{3vR}{2}$$

v = 44 m/sec

Angular momentum about B will then remain conserved

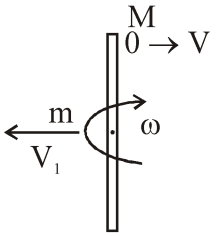
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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

27. Ans.9

Sol. (i) $L_i = L_f$; $\frac{ml^2}{12}\omega = MV \times \frac{l}{2} \Rightarrow V = \frac{m\omega l}{6M}$



(ii) $P_i = P_f$; $0 = MV - mV_1 \Rightarrow V = \frac{mV_1}{M} \Rightarrow \frac{\omega l}{6} = V_1$

(iii) $e = 1 = \frac{V-(-V_1)}{\omega\frac{l}{2}-0} \Rightarrow V + V_1 = \frac{\omega l}{2}$; $\frac{m\omega l}{6M} + \frac{\omega l}{6} = \frac{\omega l}{2}$; $\left(\frac{m}{M}+1\right) = 3$

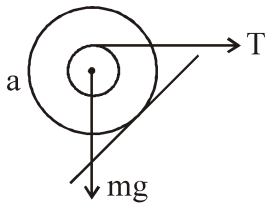
$\Rightarrow \frac{m}{M} = 2 \Rightarrow M = \frac{m}{2} = 9$ kg.

28. Ans. 4

Sol. $mv_0 \cdot 2R = \left[\frac{3}{2}mR^2 + m(2R)^2\right]\omega$; $\omega = \frac{4v_0}{11R} = 4$ rad/s

29. Ans. 2

Sol.



$mg\sin\theta - T\cos\theta = ma \therefore a = 2 m/s^2$

Subjective Type **1 Q. [4 M (0)]**

30. Ans. (i) $\frac{M\omega_0}{M+2qt}$, (ii) $q\omega_0 R^2$

Sol. (i) Since there is no external torque, conserving angular momentum about axle,

$$\frac{MR^2}{2}\omega_0 = \frac{(M+2qt)}{2}R^2\omega$$

$$\omega = \frac{M\omega_0}{(M+2qt)}$$

(ii) When angular speed is maintained

$L = I\omega_0$

w is constant, so

$$\tau = \frac{dL}{dt} = \frac{dI}{dt} \times \omega_0$$

$$I = \frac{MR^2}{2} + qtR^2$$

$$\frac{dI}{dt} = qR^2$$

$$\tau = qR^2\omega_0$$

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